Accuracy of Computed Tomography Attenuation Values in the Characterization of Pleural Fluid: An ROC Study[1]

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Rationale and Objectives. To assess the accuracy of computed tomography (CT) in characterizing pleural fluid based on attenuation values.

Materials and Methods. Protocol was approved by the local institutional review board and informed consent was waived. We retrospectively analyzed 145 pleural effusions of 145 patients (mean/standard deviation age: 60.7/15.9 years; 69 females) who underwent CT of the thorax and diagnostic thoracentesis within 7 days of each other. Effusions were classified as transudates or exudates using laboratory markers based on Light's criteria. The mean Hounsfield units (HU) of an effusion was determined by a region of interest on the three slices with the greatest anteroposterior diameter. A receiver operating characteristic curve was constructed to determine threshold values for classification on the basis of mean HU and to examine overall accuracy, using the area under the curve (A_z).

Results. Of the 101 exudates and 44 transudates, the mean attenuation of exudates (17.1 HU/standard deviation 4.4) was significantly higher than transudates (12.5 HU/6.3), ($P < .001$). There was a modest but significant positive relationship between mean HU and laboratory markers, with the strongest relationship with pleural/serum protein ($r = 0.57$, $P < .001$) and total pleural protein ($r = 0.56$, $P < .001$). The overall accuracy of attenuation values for identifying exudates was moderate, Az = 0.775, standard error = 0.039, with the largest limitation being the overlap with transudates in the 10–20 HU range, which constituted 66% (90/145) of the total effusions measured.

Conclusion. Although the mean attenuation of exudates is significantly higher than transudates, the clinical use of CT numbers to characterize pleural fluid is not recommended, as their accuracy is only moderate. Moreover, there is a notable overlap in attenuation values between transudates and exudates for a majority of effusions.

Key Words. Pleural Effusion; Tomography; X-Ray Computed; Exudates and Transudates.

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The characterization of pleural fluid is important in the clinical management of various pathologies involving the thorax, because it often provides information regarding the underlying pathologic process. Although thoracentesis is routinely used to differentiate exudates from transudate, the procedure carries small but significant risks. Even under ultrasound guidance by an interventional radiologist, there is still an approximately 1% risk of pneumothorax requiring tube thoracostomy, along with a chance for many minor complications such as pain, shortness of breath, cough, and hematoma (1). A noninvasive method to characterize pleural fluid would be valuable in guiding therapy and avoiding the potential hazards associated with thoracentesis. In addition, such a technique could possibly also help patients with a contraindication to the procedure.

The mechanism for the formation of transudates is largely dependent on an imbalance of hydrostatic and osmotic forces, and transudates generally contain low levels

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of protein and demonstrate low specific gravity (2). Exudates are usually secondary to diseases of the pleura itself and often result from lymphatic obstruction or increased capillary permeability, leading to high levels of protein and lactate dehydrogenase and elevated specific gravity. Theoretically, given the greater degree of protein content in exudates, these type of effusions could potentially demonstrate greater attenuation values on computed tomography (CT) and consequently be noninvasively differentiated from transudates.

The purpose of our study was to assess the accuracy of CT in characterizing pleural fluid based on attenuation values.

METHODS AND MATERIALS

Patient Population

This research protocol was approved by the local institutional review board and informed consent was waived. Between November 2001 and November 2004, a total of 789 patients underwent diagnostic thoracentesis. A total of 145 patients were included in our retrospective analysis based the following criteria: CT and thoracentesis performed within 7 days of each other; sufficient laboratory data for diagnosis by Light's criteria: pleural fluid lactate dehydrogenase (LDH), pleural total protein, serum LDH, and serum total protein values (3); and no history of previous pleurodesis or placement of pleural drainage catheter. Of the 644 excluded patients, 577 did not undergo diagnostic thoracentesis within 7 days of CT, 37 had insufficient laboratory information, and 30 had a history of previous intervention.

In the patient population of 145 patients with 145 laboratory analyzed effusions, there were 76 males and 69 females with a mean/standard deviation age of 60.7/15.9 years (range 3–93 years). Thoracentesis was performed a mean/standard deviation of 2.8/2.1 days within undergoing CT. Pleural LDH (U/L), pleural total protein (g/dL), pleural specific gravity, serum LDH (U/L), and serum total protein (g/dL) values were obtained in all patients, and pleural albumin (42 patients) if performed. Pleural effusions were classified based on Light's criteria, which categorizes an effusion as an exudate if it demonstrates one or more of the following features: pleural fluid protein to serum protein ratio > 0.5, pleural fluid LDH to serum LDH ration > 0.6, and pleural fluid LDH greater than two thirds the upper limit of normal serum LDH (180 U/L at our institution) (3). In addition to Light's criteria markers, the most likely etiology of the pleural fluid, such as congestive heart failure or malignancy, was determined using clinical information and a combination of serum and pleural laboratory values, such as gram stain, and cultured for bacteria, total and differential cell counts, glucose level, hematocrit, and cytologic analysis based on previously published criteria (4).



Figure 1. Contrast-enhanced axial computed tomography image of the thorax demonstrating a large left unilateral pleural effusion with a mean Hounsfield unit (HU) of 20.4 HU secondary to metastatic disease from neck squamous cell carcinoma.

Data Acquisition

CT scans were reviewed by a radiologist blinded to the clinical and laboratory information. The three slices with the greatest amount of fluid, determined by the largest anteroposterior distances of the effusion (mean for patients: 31 mm, range 11–210 mm), were used. A region of interest was placed encompassing the entire fluid on each slice and the mean Hounsfield unit (HU) value of the three slices was used (Figure 1). Care was taken not to include adjacent ribs, compressed or aerated lung parenchyma, and areas of pleural thickening. Reproducibility was assessed by an independent blinded radiologist measuring values in 50 random patients.

Scan Parameters

CT was performed on all patients on either an 8-slice scanner (LightSpeed Ultra, GE Medical Systems; Milwaukee, WI) ($n = 112$) or 16-slice scanner (LightSpeed 16, GE Medical Systems) ($n = 33$). All scans were obtained with one of the two following protocols. First, after a delay determined by an automated bolus timing program for

Table 1
Summary of Attenuation Values for Pleural Effusions

Characteristic	Number (%)	Mean/Standard Deviation Hounsfield Unit
All effusions	145/145 (100)	15.7/5.4
Exudates	101/145 (70)	17.1/4.4
Malignancy	39/145 (27)	17.3/3.6
Parapneumonic	27/145 (19)	16.9/4.9
Empyema	14/145 (10)	18.9/4.2
Pulmonary embolus	5/145 (3)	18.5/5.8
Unknown/other	16/145 (11)	14.9/4.6
Transudates	44/145 (30)	12.5/6.3
Congestive heart failure	29/145 (20)	12.7/6.9
Nephrotic syndrome	6/145 (4)	9.6/4.2
Hepatic hydrothorax	4/145 (3)	17.3/5.8
Acute atelectasis	3/145 (2)	12.0/2.9
Unknown/other	2/145 (1)	9.8/0.71

the injection of 120 mL of nonionic contrast medium at a rate of 4 mL/second, helical acquisition was performed of the thorax with the following scan parameters: 170 kV, 400–440 mA, table speed of 7.5 mm per rotation, 1.25-mm section thickness, reconstruction interval 1.25 mm, and 0.5-second gantry rotation period ($n = 34$). Second, after an empiric delay of 40 seconds for the injection of 120 mL of nonionic contrast medium at a rate of 4 mL/second, helical acquisition was performed of the thorax with the following scan parameters: 170 kV, 380–440 mA, table speed of 7.5 mm per rotation, 5.0-mm section thickness, reconstruction interval 5.0 mm, and 0.5-second gantry rotation period ($n = 111$). Twenty-eight of the scans obtained with the second protocol were performed without intravenous contrast. The image data was transferred to a computer workstation (GE Advantage Workstation 4.1) for analysis.

Statistical Analysis

Continuous data are described as the mean value ± standard deviation. The association between the mean HU of the pleural effusions and pleural LDH, pleural/serum LDH, and specific gravity was examined individually using Pearson's correlation. The relationship between attenuation and pleural total protein, pleural/serum protein, and pleural albumin was performed using linear regression. A receiver operating characteristic (ROC) curve was constructed to determine the optimal threshold value (minimal false-negative and false-positive results) to classify transudates and exudates on the basis of mean HU and to examine accuracy, using the area under the curve (A_z). Sensitivity and specificity were calculated with various cutoff points of attenuation values. Interobserver agreement was evaluated with Bland-Altman analysis (5). Statistical significance was assumed at $P < .05$. All analyses were performed using MEDCALC software program (MedCalc Software, Mariakerke, Belgium).

RESULTS

Descriptive Characteristics

Of the 145 pleural effusion, there were 101 exudates and 44 transudates by Light's criteria (3). The breakdown of the causes and their respective mean HU are summarized in Table 1. The mean attenuation of exudates (mean: 17.1/standard deviation 4.4) was significantly higher than transudates (12.5/6.3), ($P \leq .001$).

Relationship of Attenuation to Laboratory Markers

The relationship between the laboratory values and attenuation are summarized in Table 2. There was a modest but significant positive relationship between mean HU and pleural protein, pleural/serum protein, pleural albumin, pleural LDH, pleural/serum LDH, and specific gravity, with the strongest relationship with pleural protein and pleural/serum protein.

ROC Analysis

Figure 2 demonstrates the ROC curve for the accuracy of attenuation values in the identification of exudates. The overall accuracy for identifying exudates was moderate, $A_z = 0.775$, 95% confidence interval = 0.698–0.840, primarily from the overlap in scores between 10–20 HU

Table 2
Relationship of Laboratory Markers to Attenuation (HU)

Laboratory Marker	Correlation/Regression Coefficient (r/r^2)	*P* value
Pleural total protein	0.56/0.31	<.001
Pleural/serum protein	0.57/0.33	<.001
Pleural albumin	0.45/0.20	.003
Pleural lactate dehydrogenase	0.32/0.10	<.001
Pleural/serum lactate dehydrogenase	0.25/0.06	.012
Specific gravity	0.53/0.28	<.001

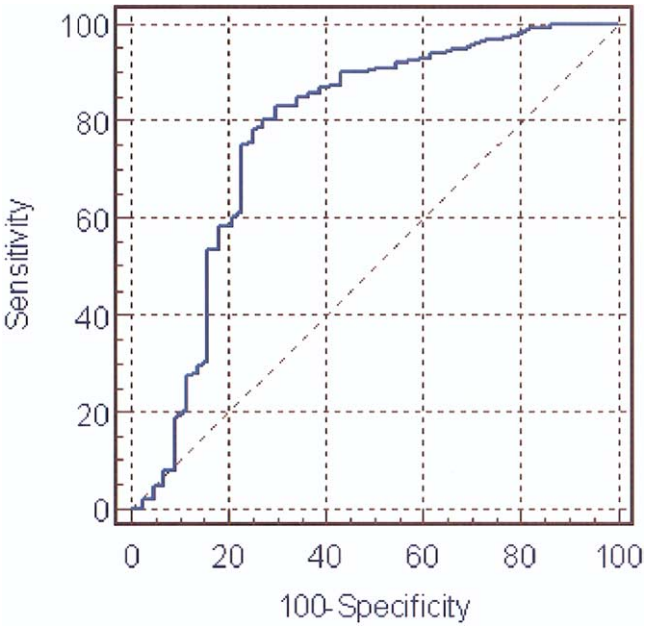


Figure 2. Receiver operating characteristic curve plotting 100-specificity (x axis) against sensitivity (y axis). Overall accuracy was moderate with area under the curve of 0.78 and standard error = 0.039.

(Figure 3), which constituted 66% (90/145) of effusions. The optimal threshold value was determined to be 13.4 HU, which demonstrated a sensitivity of 83.2% (84/101) and specificity of 70.5% (31/44) (Table 3).

Reproducibility

Figure 4 shows the plot of the differences in measurements between observer 1 and observer 2. As evident, reproducibility was good with most measurements less than 2 HU different between the observers.

Table 3
Sensitivity and Specificity for Exudates at Various HU Levels

Criterion HU	Sensitivity (95% CI)	Specificity (95% CI)
≥3.3	100.0 (96.4–100.0)	0.0 (0.0–8.1)
>5.4	100.0 (96.4–100.0)	9.1 (2.6–21.7)
>7.7	99.0 (94.6–99.8)	15.9 (6.7–30.1)
>9.1	96.0 (90.2–98.9)	29.5 (16.8–45.2)
>9.3	95.0 (88.8–98.4)	31.8 (18.6–47.6)
>9.9	94.1 (87.5–97.8)	38.6 (24.4–54.5)
>12.2	87.1 (79.0–93.0)	61.4 (45.5–75.6)
>13.4	83.2 (74.4–89.9)	70.5 (54.8–83.2)
>15.1	64.4 (54.2–73.6)	77.3 (62.2–88.5)
>17.1	43.6 (33.7–53.8)	84.1 (69.9–93.3)
>18.5	33.7 (24.6–43.8)	84.1 (69.9–93.3)
> 20.3	27.7 (19.3–37.5)	88.6 (75.4–96.2)
> 22.6	9.9 (4.9–17.5)	90.9 (78.3–97.4)
>24.4	4.0 (1.1–9.8)	95.5 (84.5–99.3)
>26.4	2.0 (0.3–7.0)	95.5 (84.5–99.3)
>28.3	2.0 (0.3–7.0)	97.7 (87.9–99.6)

HU: Hounsfield unit; CI: confidence interval.

DISCUSSION

Pleural effusions affect 1.3 million patients annually in the United States, mostly secondary to congestive heart

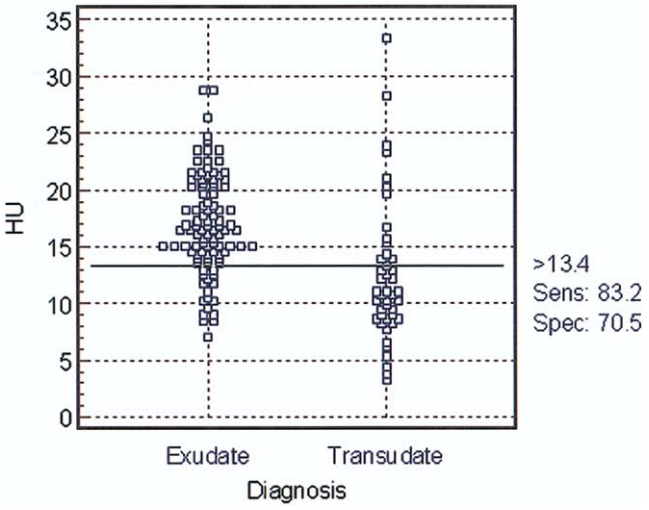


Figure 3. Interactive dot diagram plotting the attenuation values on the y axis for exudates and transudates. A level of 13.4 Hounsfield units (HU) was the optimal threshold value (minimal false-negative and false-positive results), with a sensitivity of 83.2% (84/101) and specificity of 70.5% (31/44).

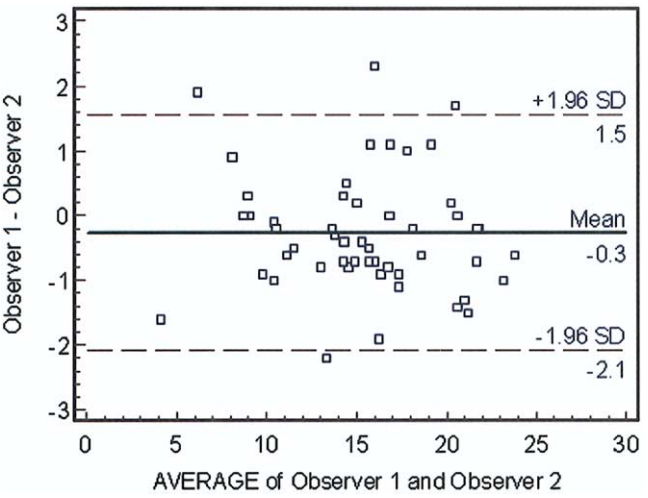


Figure 4. Bland-Altman plot with the differences between the observers on the y axis and average of the observers on the x axis. Most measurements between the observers were less than 2 Hounsfield unit different.

failure (CHF) (2). Although many do not require further characterization based on clinical suspicion, thoracentesis is usually indicated for the presence of an effusion of unknown cause greater than 10-mm thick on ultrasound or decubitus radiography, or in patients with CHF with an asymmetric effusion, fever, or chest pain (6–9). Although thoracentesis is generally considered a relatively benign procedure, there are associated risks and contraindications, such as coagulopathy. A noninvasive method to help direct the clinical pathway could reduce the number of unnecessary procedures. However, our study found that CT likely has no significant role in stratifying pleural effusions as exudates or transudates based on attenuation. Exudates exhibited significantly higher CT number than transudates, as there was a significant albeit modest positive correlation between HU and the laboratory markers of Light's criteria (3). However, the overall accuracy of CT attenuation values was only moderate ($A_z = 0.78$), and there was considerable overlap in values for transudates and exudates for a majority of effusions in the 10–20 HU range, 66% (90/145).

Few studies have been published on the characterization of pleural abnormalities using noninvasive modalities such as CT. Aquino et al examined 86 patients with contrast-enhanced CT and found that parietal pleural thickening was almost invariably an indicator of an exudate, and that transudates were rarely associated with the finding (10). Leung et al studied 74 patients and demonstrated that circumferential pleural thickening, nodular pleural thickening, parietal pleural thickening greater than 1 cm, and mediastinal pleural involvement are sensitive but relatively nonspecific markers of malignant pleural disease (11). Previous studies have not assessed the accuracy of attenuation values in assessing pleural fluid. Our study demonstrates limited potential for the utilization of CT attenuation measurements in evaluating effusions. In our experience, many radiologists may incorrectly classify effusions as "simple" or "complex" with unsubstantiated numerical thresholds for differentiation. Our findings suggest that their clinical use should be discouraged.

There are multiple limitations to our study. First of all, although Light's criteria are considered the gold standard for characterizing pleural effusion, they are less accurate in the setting of patients with CHF who are on diuresis, where transudates can be misclassified as exudates (12,13). However, they remain the best method of differentiating transudates and exudates (14). Next, there were no patients with hemothorax or chylothorax in our study. Consequently, our data cannot be applied to these populations. Another limitation is that two different scanning parameters were used, and some patients received intravenous contrast, whereas others did not. Although there were no noticeable differences by either radiologist, future studies may want to employ a standard technique, preferably noncontrast to decrease potential volume averaging with adjacent high-attenuation structures, to see if this would improve the accuracy of CT numbers. Finally, we determined the mean HU using only the three axial slices with the greatest amount of fluid determined by the anteroposterior distance. The use of all slices to determine the mean may result in variations from our data. Moreover, there is a relative bias in our study toward larger effusions, because patients with smaller effusions (usually <10 mm) do not undergo thoracentesis. It is likely that volume averaging would play a larger role in smaller effusions making attenuation values less accurate.

In conclusion, attenuation values demonstrate little potential clinical value in the characterization of pleural fluid, secondary to limited accuracy and an overlap in values in a majority of effusions (7–9).

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